

TITLE: GXP and Non-GXP Study Report Review and Approval Form

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Study Title: Efficacy of B3 antibody in a mouse model of Collagen Antibody Induced Arthritis (CAIA). Wash-out study			
Study Vendor/Site: ☐ N/A Karolinska Inst.	Study Director: ☐ N/A Camilla Svensson	Study Contributor(s): ☐ N/A Ann Gregus	
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Executive summary of study results: ☐ See Attached ☐ N/A Arthritis was induced in male CBA mice by i.v. injection of a cocktail of anti-collagen antibodies on Day 0 followed by i.p. injection of 25 ug of lipopolysaccharide (LPS) on Day 5. Starting on Day 12 at the height of the inflammatory response, animals (6 mice/group) were treated twice weekly with Control IgG or anti-LPS antibody, B3, 2 or 10 mg/kg, s.c., treatments continued to day 43 when antibody treatment was halted. On day 57, CAIA animals that had previously received only saline were switched to receive twice weekly B3, 10 mg/kg; and animals receiving B3 10 mg/kg were switched to receive 10 mg/kg IgG control. Mechanical thresholds and the degree of arthritis were assessed with von Frey filaments and visual scoring, respectively. Results showed that treatment with 10 mg/kg B3 resulted in a decrease in the degree of allodynia and that B3 at 2 mg/kg appeared to be partially effective whereas, control antibody had no effect. When treatment of the 10 mg/kg group was stopped at Day 43 a gradual decrease in protection was seen that paralleled the expected time course for loss of B3 from the circulation. On the other hand, CAIA animals that had previously been treated with saline were switched to 10mg/kg of B3 twice weekly starting on day 57, when mechanical hypersensitivity was well established and results showed that B3 was also efficacious when treatment was delayed. In summary, anti-LPS antibody decreased tactile hypersensitivity in arthritic mice in a dose dependent fashion and B3 therapy was highly effective when delayed until mechanical hypersensitivity was well established.			
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Report Number: LT3114-PHA-005-R	
Title: Efficacy of B3 antibody in a mouse model of Collagen Antibody Induced Arthritis (CAIA). Wash-out study	Page 1 of 9

Efficacy of B3 antibody in a mouse model of Collagen Antibody Induced Arthritis (CAIA). Wash-out Study

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TABLE OF CONTENTS

1	SUMMARY	3
2	INTRODUCTION	3
3	MATERIALS AND METHODS.....	4
3.1	Test Article and Control.....	4
3.2	Study Design.....	4
3.3	Experimental outline.....	5
3.4	Tactile Response Threshold.....	5
3.5	Clinical scores of arthritis	6
3.6	Data Analyses	6
4	RESULTS	6
5	CONCLUSIONS.....	8
6	DATA ARCHIVE.....	9

LIST OF IN-TEXT FIGURES

Figure 1.	Anti-LPA antibody attenuates CAIA-induced hypersensitivity	7
Figure 2.	Tactile alloynia is re-established in CAIA following LPA Ab (10 mg/kg) wash-out.	8

1 SUMMARY

This efficacy study in the collagen antibody-induced arthritis (CAIA) model was conducted at the Department of Physiology and Pharmacology and Neuroscience, at the Karolinska Institutet, (Stockholm, Sweden). The purpose of the study was to test the effect of a two doses of a murine anti-LPA antibody (B3), to examine the duration of protection once treatment was stopped, and to determine whether B3 could be effective when the start of treatment was delayed until all inflammation had resolved.

Arthritis was induced in male CBA mice by i.v. injection of a cocktail of anti-collagen antibodies on Day 0 followed by i.p. injection of 25 µg of lipopolysaccharide (LPS) on Day 5. Starting on D12 at the height of the inflammatory response, animals (6 mice/group) were treated twice weekly with control IgG or B3, 2 or 10 mg/kg, s.c., treatments continued to D43 when antibody treatment was halted. On D57, CAIA animals that had previously received only saline were switched to receive twice weekly B3, 10 mg/kg; and animals receiving B3 10 mg/Kg were switched to receive 10 mg/Kg IgG control. Mechanical thresholds and the degree of arthritis were assessed with von Frey filaments and visual scoring, respectively. Results showed that treatment with 10 mg/kg B3 resulted in a decrease in the degree of allodynia and that B3 at 2 mg/kg appeared to be partially effective whereas, control antibody had no effect on tactile hypersensitivity. When treatment of the 10 mg/kg group was stopped at D43 a gradual decrease in protection was seen that paralleled the expected time course for loss of B3 from the circulation. On the other hand, CAIA animals that had previously been treated with saline were switched to receive 10 mg/kg of B3 twice weekly starting on D57, when mechanical hypersensitivity was well established and results showed that B3 was also efficacious when treatment was delayed.

In summary, a murine anti-LPA antibody (B3) decreased tactile hypersensitivity in arthritic mice in a dose dependent fashion and B3 therapy was highly effective when delayed until mechanical hypersensitivity was well established and all signs of inflammation had disappeared in the disease model.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is a key extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA Ab). In preliminary studies, Lpath's anti-LPA mAb has demonstrated efficacy in the Seltzer model of neuropathic pain where the literature has demonstrated a role for LPA in the disorder, we have also demonstrated efficacy of the anti-LPA antibody in ameliorating neuropathic pain in animals models of type 1 and 2 diabetes. We next wanted to expand the efficacy studies to explore the effect of the murine, B3, in a model of chronic arthritis pain. The pain that

accompanies chronic rheumatoid arthritis (RA) is characterized as having both inflammatory and non-inflammatory components and this non-inflammatory component of pain shares many aspects of the neuropathic pain seen in other disease states. This pain can be reproduced in mouse models of RA including the K/BxN serum transfer model and the collagen antibody induced arthritis (CAIA) model. This study was performed in the laboratory of Dr Camilla Svensson at Karolinska Institute. Dr. Svensson has recently characterized the collagen antibody-induced arthritis (CAIA) model using pain readouts beyond the inflammatory phase of the disease and established that the antibody-induced arthritis generates persistent pain-like behavior that outlasts joint inflammation by weeks, a characteristic that mimics the human disease. Thus, studies were undertaken to determine the efficacy of anti-LPA antibody treatment in relieving chronic non-inflammatory pain in the CAIA model. In a previous study it was found that B3 treatment decreased tactile hypersensitivity but at a higher dose of antibody (40 mg/Kg) no efficacy was seen. Thus, this study was designed to test the effect of two doses (2 and 10 mg/Kg) of antibody, to examine the duration of the protection once treatment was stopped, and to determine whether antibody could be effective when the start of treatment was delayed until all inflammation had resolved i.e. to demonstrate a therapeutic effect of antibody treatment that was not solely as a result of blocking the onset of allodynia in this model.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Murine anti-LPA antibody (504B3) Lot# HR101201

Murine IgG2b antibody (LT1015) Lot# HR91184

3.2 Study Design

Animals were habituated to the vivarium for 7 days after arrival. Over the following 9 days the mice were habituated to the experimental personnel and the testing devices. Baseline allodynia were collected every third day for three consecutive testing times prior to injection of the anti-collagen antibody cocktail.

Starting on D12 at the height of the inflammatory response, animals (6 mice/group) were treated twice weekly with control IgG (LT1015) or B3 antibody at 2 or 10 mg/kg, s.c. Control groups received saline only (no CAIA, control), or anti-collagen plus LPS with saline twice weekly (CAIA)

Six groups of male CBA mice (total 36 mice) were included in the study. Five of the groups were subjected to Collagen Antibody Induced Arthritis (CAIA) by i.v. injection of a cocktail of anti-collagen antibodies on Day 0 of the study, followed by i.p. injection of 25 µg of lipopolysaccharide (LPS) on Day 5. Mechanical thresholds and the degree of arthritis were assessed as described below.

3.3 Experimental outline

Day -9, -6 and -3: Baseline allodynia and arthritis scores

Day 0: Intravenous injection of collagen antibody cocktail (1.5 mg/mouse)

Day 5: Intraperitoneal injection of 25 µg LPS to boost arthritis

Day 6 and 9: Allodynia and arthritis scores were assessed

Subcutaneous (s.c.) injection of B3 or LT1015 on day **12, 15, 18, 22, 25, 29, 32, 36, 39, 43**

Allodynia and arthritis scores assessed day **12** (prior to injection) **13, 16, 19, 21, 23, 30, 33, 37, 40, 44**

Drug wash out from day 40 to 72 (initial CAIA + B3 Ab at 10mg/kg group switched to saline)

Allodynia and arthritis score measurement day **44, 51, 54, 57, 58, 61, 65, 68 and 71**

Drug cross-over from day 58 to 72 (initial CAIA + saline group switched to B3 Ab 10mg/kg) s.c injection of B3 or LT1015 day 57, 60, 64, 67, and 70.

Allodynia and arthritis scores were assessed on day **58, 61, 65, 68, 71**

Day 72: **End of Study**

3.4 Tactile Response Threshold

Mice were transferred to a testing cage with a wire mesh bottom and allowed to acclimate and tactile allodynia was assessed using von Frey filaments and the “up-down method” for calculation of 50% withdrawal threshold. On days where assessment of allodynia coincided with antibody treatment, the measurement of allodynia was performed first.

3.5 Clinical scores of arthritis

Arthritic clinical scores were assessed at the same time as the allodynia and grip strength measures were made. The degree of arthritis was assessed by lifting the animal and inspecting the digits, dorsal aspect of the paws and the ankle joints. Swelling was evaluated according to the scoring system below. Redness was not considered.

Digits: 0-5 points (1 point for each swollen digit)

Paw: 0 or 5

Ankle: 0 or 5

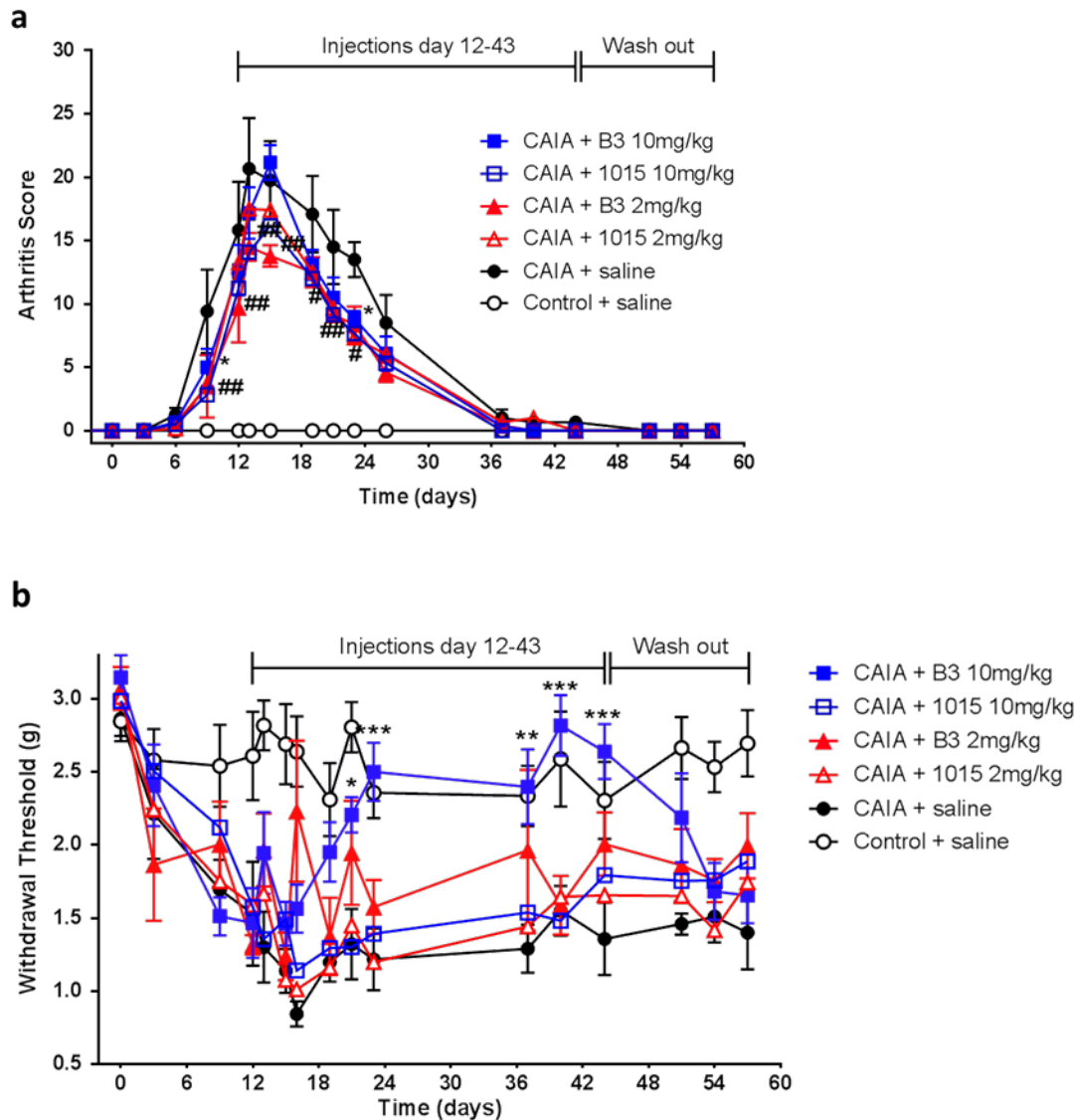
The maximal score per leg is 15 points. All four legs were scored, hence the maximal score per mouse was 60 points.

3.6 Data Analyses

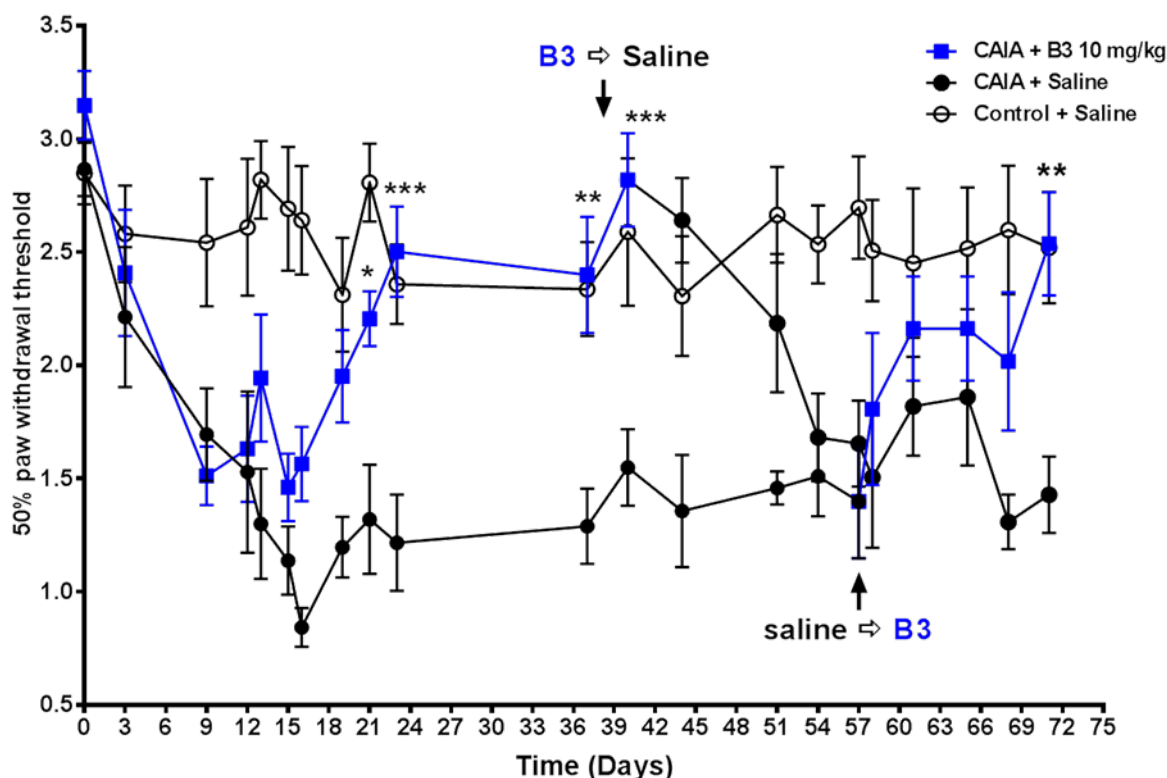
All data was generated and collected at the laboratory of Dr. Svensson at Karolinska Institute. Descriptive statistics was used to calculate median, mean, standard deviation (SD) and standard error of the mean (SEM) using Microsoft Excel. An excel spread sheet containing all data and analysis was sent via e-mail to Lpath. GraphPad Prism software (version 6.0, Graph Software inc, La Jolla, CA) was used to graph and analyze (ANOVA) the data.

4 RESULTS

Treatment with B3 resulted in reduction of tactile hypersensitivity (Figure 1b.) with a no specific, small effect on arthritis score at the lower dose of antibody treatment (Figure 1a.). No significant differences were observed between CAIA Saline and 1015 treated animals (Figure 1b). After 12 days of treatment with 10 mg/kg, animals receiving B3 were indistinguishable from “no CAIA” control animals. B3 at 2 mg/kg was partially effective whereas control antibody had no effect on tactile hypersensitivity (Figure 1). When treatment of the 10 mg/kg group was stopped at D43 a gradual decrease in protection was seen that paralleled the expected time course for loss of B3 from circulation (blue line in Figure 1 and Figure 2). CAIA animals that had previously been treated with saline (black line in Figure 1 and Figure 2) were switched to receive 10 mg/kg of B3 twice weekly starting on D57 to examine the effect on established allodynia. As can be seen in Figure 2, a progressive decrease in tactile hypersensitivity was seen in these animals until, after 2 weeks of treatment, this group was indistinguishable from non-arthritic control animals. Thus, treatment with B3 was highly effective even when delayed until mechanical hypersensitivity was well established in this disease model.

Figure 1. Anti-LPA antibody attenuates CAIA-induced hypersensitivity

Graphs depicting arthritis scores (a) and tactile allodynia (b) in control mice, and CAIA mice that received saline, B3 or 1015 (2 and 10 mg/kg). Arthritis scores were assessed by visual inspection, max score was 60 per mouse. Each time point represents mean $\pm$ SEM. ## p < 0.05, ### p < 0.01 v. CAIA + 2 mg/kg B3 v CAIA + Saline and ** p < 0.05, *** p < 0.01, *** p < 0.001 CAIA + 10 mg/Kg B3 v. CAIA + Saline by two-way repeated measures ANOVA with Bonferroni post-hoc analysis.

Figure 2. Tactile allodynia is re-established in CAIA following LPA Ab (10 mg/kg) wash-out.

Mechanical hypersensitivity (tactile allodynia) was assessed by Von Frey filaments according to the “up-down method, and the data processed to calculate 50% probability to get a withdrawal response. Control (black diamond) and CAIA (allodynic) mice were treated s.c, starting on day 12 with either saline (Saline group, red triangle) or 10mg/kg B3 mAb (blue circle) two times a week (Figure 1). Antibody treatment was switched to saline treatment on day 43 for the mAb group and allodynia was followed to the end of the study (day 72). The saline group was switched to antibody treatment (10 mg/Kg, twice a week) on day 57 to day 72. Each time point represents mean $\pm$ SEM, n=6 mice/group. **p < 0.05, ***p < 0.01, ****p < 0.001 CAIA + 10 mg/Kg B3 v. CAIA + Saline by two-way repeated measures ANOVA with Bonferroni post-hoc analysis.

5 CONCLUSIONS

The anti-LPA antibody (B3) decreased tactile hypersensitivity in arthritic mice in a dose dependent fashion.

The reduction in hypersensitivity persisted after treatment was stopped and the duration of this was consistent with the expected elimination of the antibody.

Not only was B3 efficacious when treatment was started early in the disease, but the treatment was also highly effective when delayed until mechanical hypersensitivity was well established and all signs of inflammation had disappeared.

Report Number: LT3114-PHA-005-R	
Title: Efficacy of B3 antibody in a mouse model of Collagen Antibody Induced Arthritis (CAIA). Wash-out study	Page 9 of 9

6 DATA ARCHIVE

Lpath Notebook #137 pp. 64-82